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2026
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SSC

— The Complete Book —

MATHEMATICS

Revision | Practice | PYQS

Chapter-Wise (Type wise) 3000+MCQs

With Exam hall approach solutions



Useful for

SSC CGL, CHSL (PRE + MAINS),
RAILWAY, AND OTHER GOVT. EXAMS

Each chapter is divided
into 4 smart parts



Quick Revision



Type-wise Practice Set



Mixed Practice Set



Hard Ques Practice



Abhishek Sir
(Maths Mania)



Also includes:

Complete Advanced Maths Formula

≡ HOW TO USE THIS BOOK ≡

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This book is designed to help you **learn**, **revise**, **practice**, and **master** every chapter in the most efficient way possible. To get the maximum benefit, follow the sequence below:

1

LEARN

Scan the QR Code given with each chapter to access the complete video lecture. The lectures cover all concepts and important question types in a comprehensive manner.



SCAN ME

2

QUICK REVISION - COMPLETE CHAPTER REVISION

After completing the lecture, go through the Quick Revision section to revise the entire chapter and recall all important concepts and types.



3

SET-01 - TYPE-WISE PRACTICE

Solve Set-1 first. It contains carefully selected questions covering all important concepts and question types of the chapter. This is the most important practice set.



4

SET-02 - MIXED PRACTICE

Once you have completed Set-1 for all chapters, attempt Set-2 for extra practice, better accuracy, and improved speed.



5

SET-03 - HARD QUES PRACTICE

If you want to challenge yourself further, solve Set-3, which contains advanced-level problems requiring deeper understanding and application.



★ SUCCESS FORMULA ★



LEARN



REVISE



PRACTICE



IMPROVE



MASTER

LEARN

QUICK REVISION

SET-01
TYPE-WISE PRACTICE

SET-02
MIXED PRACTICE

SET-03
HARD QUES PRACTICE

MASTERY





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(Maths Mania)

Co-Author & Editor

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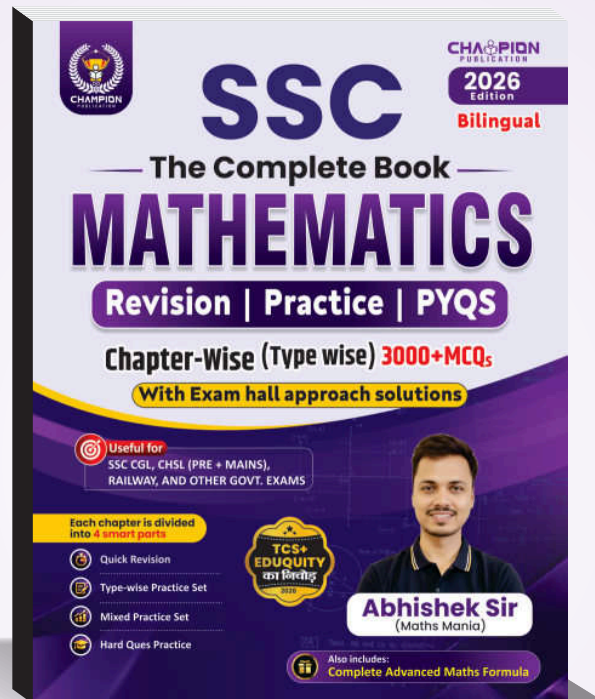
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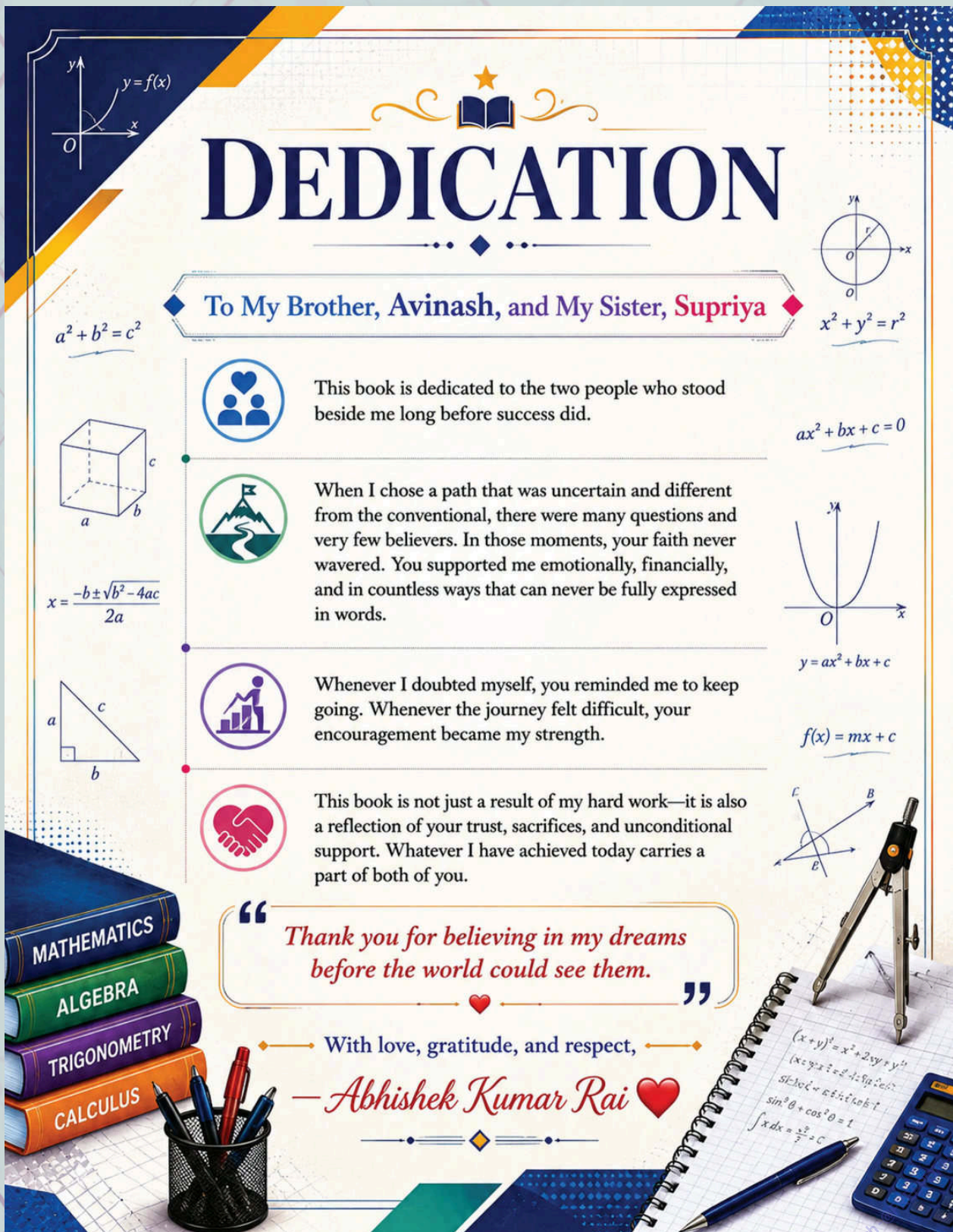
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PROGRESS REPORT											
Chapter Name	Lecture	Revision				Set-01		Set-02		Set-03	
Percentage											
Profit & Loss											
Discount											
Simple Interest											
Compound Interest											
SI & CI Installment											
Ratio & Proportion											
Age											
Partnership											
Average											
Mixture & Alligation											
Time & Work											
Work & Wages											
Pipe & Cistern											
Time, Speed & Distance											
Boat & Stream											
Race											
Number System											
Sequence & Series											
Simplification											
Surds and Indices											
LCM & HCF											
Geometry											
Polygon											
Algebra											
Polynomials											
Trigonometry											
Height & Distance											
2D-Mensuration											
3D-Mensuration											
Co-ordinate Geometry											
Permutation and Combination											
Probability											
Statistics											
Data Interpretation											

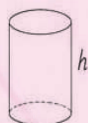
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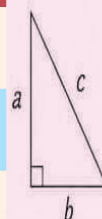
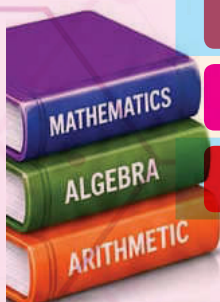
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$$a^2 + b^2 = c^2$$



$$V = \pi r^2 h$$



$$a^2 + b^2 = c^2$$



$$V = a^3$$



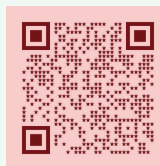
$$A = \pi r^2$$



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GEOMETRY (ज्यामिति)

(Lines & Angles, Circles, Triangles, Quadrilaterals)



Formula Playlist

LINES & ANGLES (रेखाएँ और कोण)

Zero Angle / शून्य कोण
($\theta = 0^\circ$)

Acute Angle / न्यून कोण
($0^\circ < \theta < 90^\circ$)

Right Angle / समकोण कोण
($\theta = 90^\circ$)

Obtuse Angle / अधिक कोण
($90^\circ < \theta < 180^\circ$)

Straight Angle / ऋजु कोण/रेखीय कोण
($\theta = 180^\circ$)

Reflex Angle / बृहत्कोण/प्रतिवर्ती कोण
($180^\circ < \theta < 360^\circ$)

Complete Angle / सम्पूर्ण कोण
($\theta = 360^\circ$)

If $\angle a + \angle b = 90^\circ$
then $\angle a$ and $\angle b$ are,
Complementary Angles (पूरक कोण)

If $\angle a + \angle b = 180^\circ$
then $\angle a$ and $\angle b$ are,
Supplementary Angles (सम्पूरक कोण)

If $\angle a + \angle b = 360^\circ$
then $\angle a$ and $\angle b$ are,
Conjugate angles (संयुग्मी कोण)

If $l \parallel m$ and
 t is a transversal line (अनुप्रस्थ रेखा)

Note:
 $\angle 3 + \angle 6 = \angle 4 + \angle 5 = 180^\circ$
 (sum of interior angles of same side)
 $\angle 1 + \angle 8 = \angle 2 + \angle 7 = 180^\circ$
 (sum of exterior angles of same side)

Corresponding Angles / संगत कोण
 $\angle 1 = \angle 5, \angle 2 = \angle 6, \angle 3 = \angle 7, \angle 4 = \angle 8$

Alternate Interior Angles / एकांतर आंतरिक कोण
 $\angle 3 = \angle 5, \angle 4 = \angle 6$

Alternate Exterior Angles / एकांतर बाह्य कोण
 $\angle 1 = \angle 7, \angle 2 = \angle 8$

Vertically Opposite Angles / शीर्षाभिमुख कोण
 $\angle 1 = \angle 3, \angle 2 = \angle 4, \angle 5 = \angle 7, \angle 6 = \angle 8$

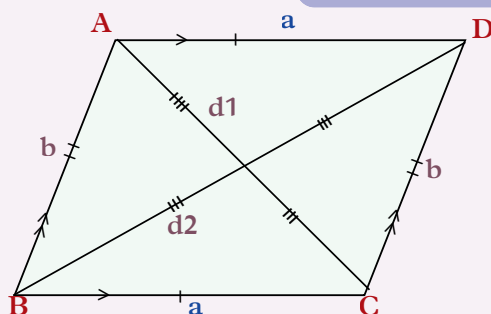
If $l \parallel m$
 $x + y + z = \alpha + \beta + \theta$

If $l \parallel m$
 $x = \alpha + \beta + \theta$

If $l \parallel m$
 $\alpha + \beta + \gamma = 360^\circ$

QUADRILATERALS (चतुर्भुज)

PARALLELOGRAM (समांतर चतुर्भुज)



- ♦ Opposite sides are parallel and equal.
- ♦ Opposite angles are equal.
- ♦ Any two adjacent or consecutive angles are supplementary—their sum is 180° .
- ♦ Diagonals bisect each other. A diagonal divides it into two congruent triangles.

- ♦ विपरीत भुजाएँ समान और समांतर होती हैं।
- ♦ विपरीत कोण समान होते हैं।
- ♦ किसी भी दो क्रमागत अथवा सन्निकट कोणों का योग 180° होता है।
- ♦ विकर्ण एक-दूसरे को समद्विभाजित करते हैं। एक विकर्ण इसे दो सर्वांगसम त्रिभुजों में विभाजित करता है।

Parallelogram Law

$$(AC)^2 + (BD)^2 = 2(AB)^2 + 2(BC)^2$$

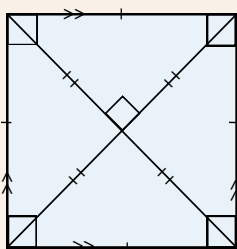
Sum of square of diagonals = Sum of square of sides

Note:

Every square, rectangle, and rhombus is a parallelogram, but every parallelogram is not necessarily a square, rectangle, or rhombus.

हर वर्ग, आयत और समचतुर्भुज एक समांतर चतुर्भुज होता है, लेकिन हर समांतर चतुर्भुज का वर्ग, आयत या समचतुर्भुज होना ज़रूरी नहीं है।

SQUARE (वर्ग)



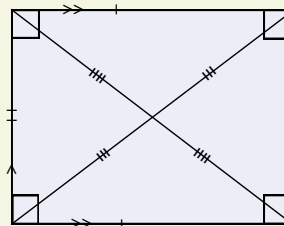
- ♦ All sides are equal.
- ♦ Opposite sides are parallel.
- ♦ All angles are equal to 90° .
- ♦ Diagonals bisect each other at right angles and are equal.

All squares are rectangles. But all rectangles are not squares.

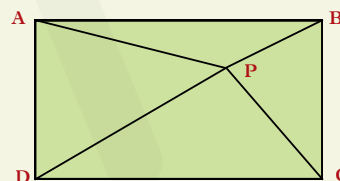
- ♦ सभी भुजाएँ बराबर होती हैं।
- ♦ विपरीत भुजाएँ समांतर होती हैं।
- ♦ सभी कोण 90° के होते हैं।
- ♦ विकर्ण एक-दूसरे को समकोण पर बराबर भागों में विभाजित करते हैं।

RECTANGLE (आयत)

- ♦ Opposite sides are parallel.
- ♦ Opposite sides are equal.
- ♦ All angles are equal to 90° .
- ♦ Diagonals bisect each other and are equal.



- ♦ विपरीत भुजाएँ समांतर होती हैं।
- ♦ विपरीत भुजाएँ बराबर होती हैं।
- ♦ सभी कोण 90° के होते हैं।
- ♦ विकर्ण एक-दूसरे को बराबर भागों में विभाजित करते हैं और बराबर होते हैं।

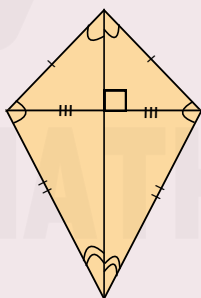


In a rectangle, if P is any point inside the rectangle,

$$PA^2 + PC^2 = PB^2 + PD^2$$

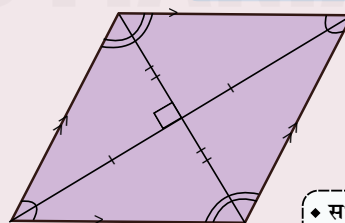
KITE (पतंगाकार चतुर्भुज)

- ♦ Has two pairs of adjacent equal sides.
- ♦ One pair of opposite angles is equal.
- ♦ Diagonals are perpendicular, and the longer diagonal bisects the shorter one.
- ♦ Longer diagonal divides kite into two congruent triangles.



- ♦ दो जोड़ी आसन्न भुजाएँ बराबर होती हैं।
- ♦ एक जोड़ी विपरीत कोण बराबर होते हैं।
- ♦ विकर्ण लम्बवत होते हैं और बड़ा विकर्ण छोटे विकर्ण को बराबर भागों में विभाजित करता है।
- ♦ बड़ा विकर्ण पतंग को दो सर्वांगसम त्रिभुजों में विभाजित करता है।

RHOMBUS (समचतुर्भुज)



- ♦ All sides are equal.
- ♦ Opposite sides are parallel.
- ♦ Opposite angles are equal.
- ♦ Diagonals bisect each other perpendicularly.

- ♦ सभी भुजाएँ बराबर होती हैं।
- ♦ विपरीत भुजाएँ समांतर होती हैं।
- ♦ विपरीत कोण बराबर होते हैं।
- ♦ विकर्ण एक-दूसरे को समकोण पर बराबर भागों में विभाजित करते हैं।

TRAPEZIUM (समलंब चतुर्भुज)

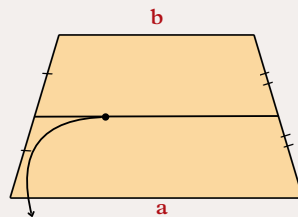


Exactly one pair of opposite sides is parallel.

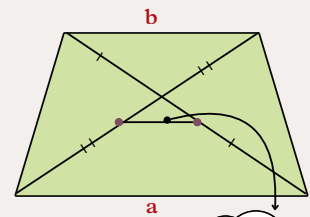
केवल एक जोड़ी विपरीत भुजाएँ समांतर होती हैं।



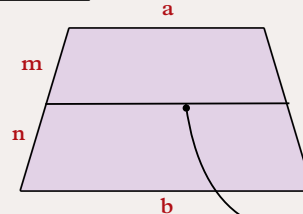
Non-parallel sides are equal :-
Isosceles Trapezium.



$$\frac{1}{2}(a+b)$$

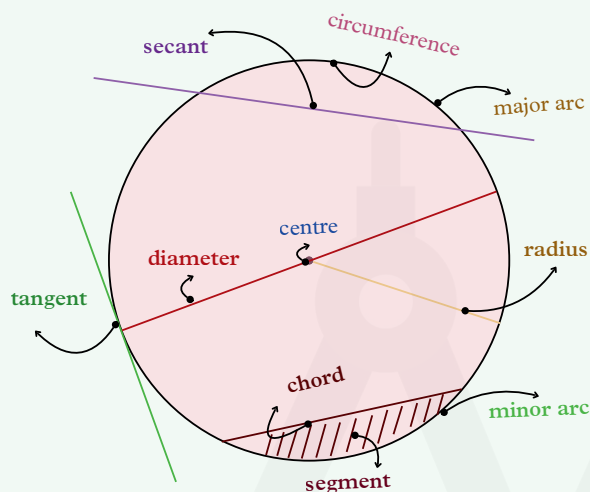


$$\frac{1}{2}|a-b|$$



$$\frac{m \cdot b + n \cdot a}{m+n}$$

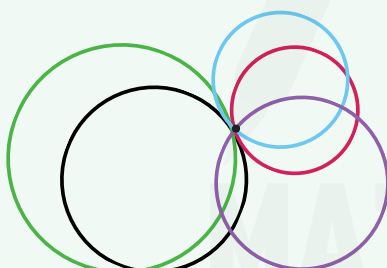
CIRCLE (वृत्त)



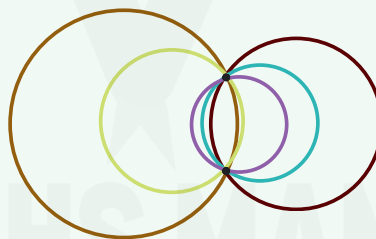
- Circle - वृत्त
- Center - केंद्र
- Radius - त्रिज्या
- Diameter - व्यास
- Chord - जीवा

- Secant - छेदक रेखा
- Segment - वृत्तखंड
- Minor Arc - लघु चाप
- Major Arc - दीर्घ चाप
- Tangent - स्पर्श रेखा

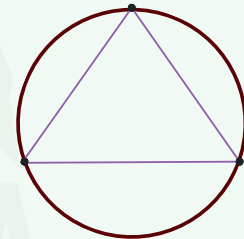
CIRCLES THROUGH GIVEN POINTS (दिए गए बिंदुओं से होकर जाने वाले वृत्त)



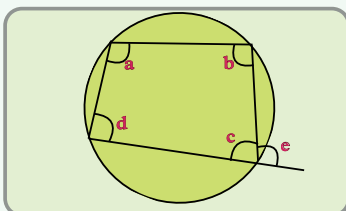
ONE point = infinite circles



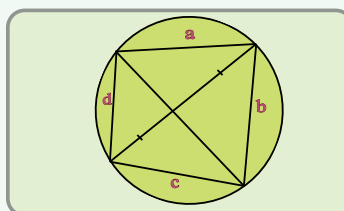
TWO points = infinite circles



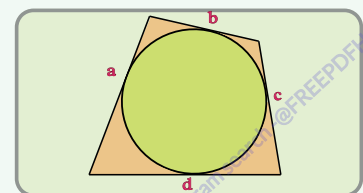
THREE non-collinear points = one circle



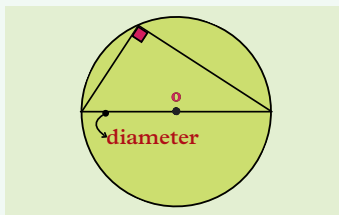
In cyclic quadrilateral (चक्रीय चतुर्भुज)
 $a+c = b+d = 180^\circ$
 $a = e$



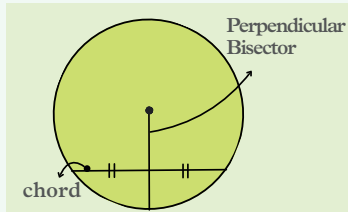
When one diagonal of cyclic quadrilateral
bisects another diagonal
 $a \times b = c \times d$



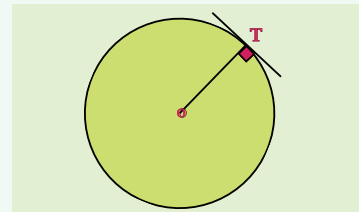
In a tangential quadrilateral
(स्पर्शीय चतुर्भुज) the sums of opposite sides
are equal.
 $a + c = b + d$



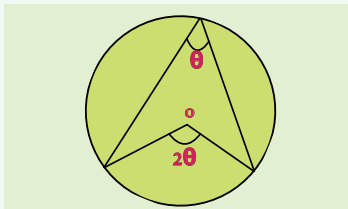
An angle subtended by a diameter at the circle is 90° .



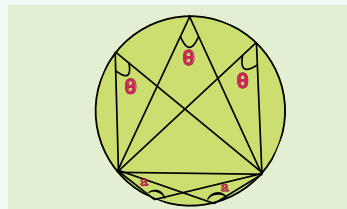
The perpendicular bisector of a chord passes through the centre of the circle.



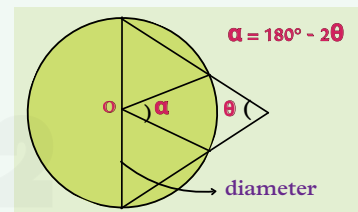
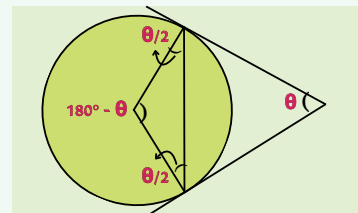
The radius at the point of contact is perpendicular to the tangent.



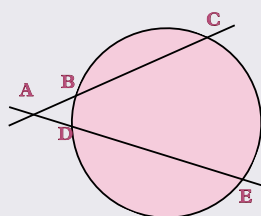
The angle subtended by an arc at the centre is twice the angle subtended at the circumference.



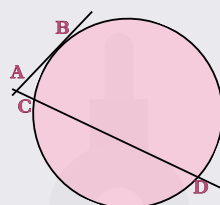
Angles subtended by the same arc at the circumference are equal.



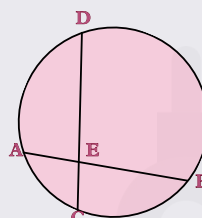
TANGENT-SECANT THEOREM (स्पर्शिका-छेदक प्रमेय)



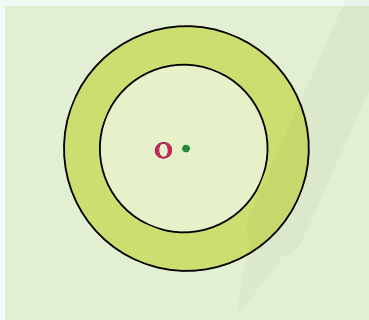
$$AB \times AC = AD \times AE$$



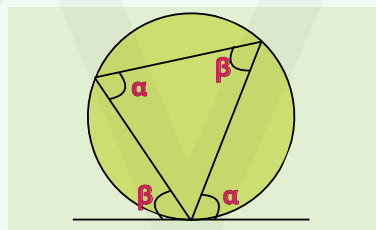
$$AB^2 = AC \times AD$$



$$AE \times EB = DE \times EC$$

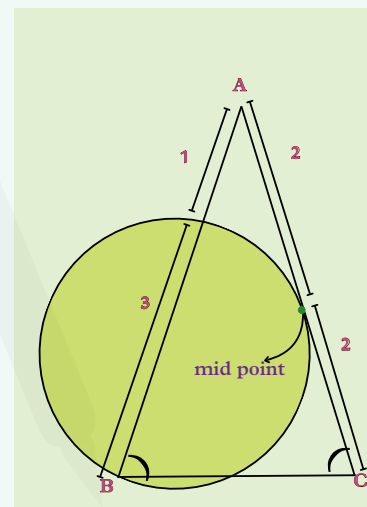


Concentric Circles
(सहकेन्द्रीय वृत्त)



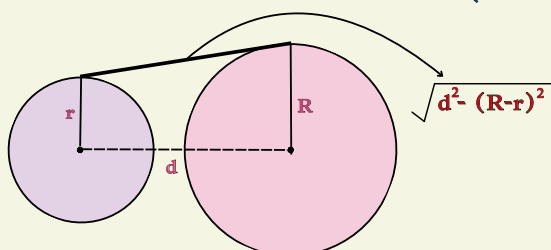
Alternate Segment Theorem
(वैकल्पिक खंड प्रमेय)

The angle between a tangent and a chord at the point of contact equals the angle in the alternate segment.

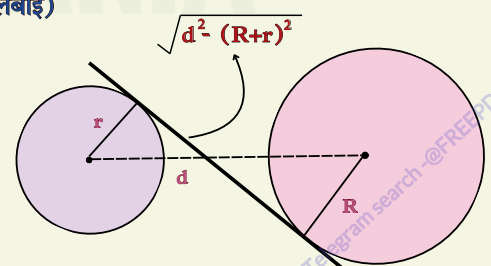


ABC is an isosceles triangle
(समद्विबाहु त्रिभुज)

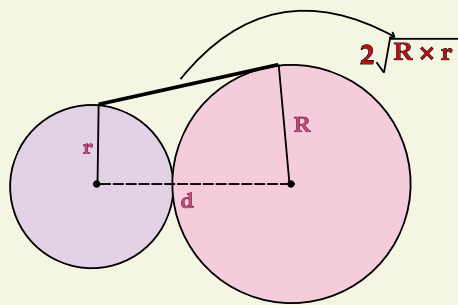
LENGTH OF COMMON TANGENTS (समान स्पर्शिकाओं की लंबाई)



DIRECT/EXTERNAL COMMON TANGENT
स्पर्शिका प्रमेय

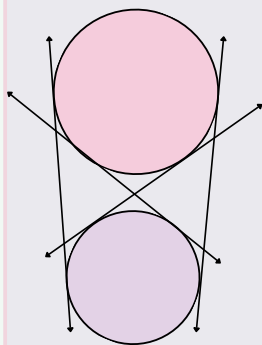


TRANSVERSE / INTERNAL COMMON TANGENT
अनुप्रस्थ स्पर्शिका प्रमेय

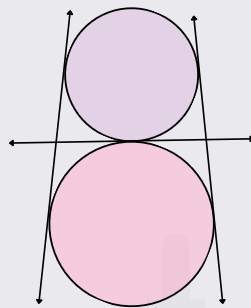


CIRCLE TOUCHING EXTERNALLY (बाहर से स्पर्श करते हुए वृत्त)

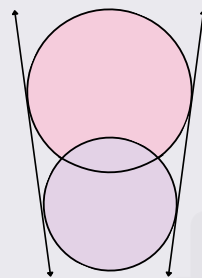
COMMON TANGENTS OF TWO CIRCLES (दो वृत्तों की समान स्पर्शिकाएँ)



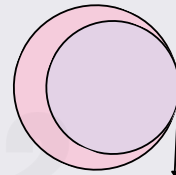
$d > r_1 + r_2$
2 Direct CT
2 Transverse CT



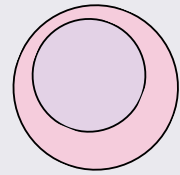
$d = r_1 + r_2$
2 Direct CT
1 Transverse CT



$r_1 - r_2 < d < r_1 + r_2$
2 Direct CT

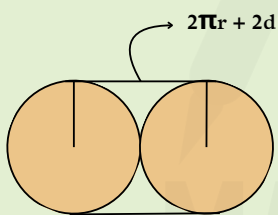


$d = r_1 - r_2$
1 Direct CT

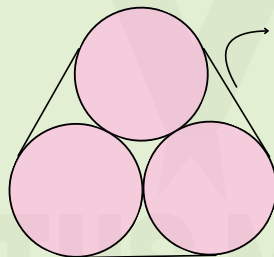


$d < r_1 - r_2$
No CT

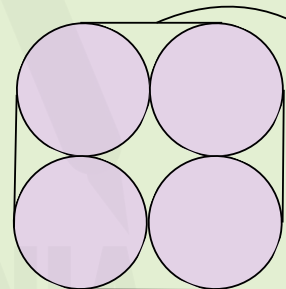
LENGTH OF STRING WRAPPED AROUND EQUAL CIRCLES (समान वृत्तों के चारों ओर लिपटी डोरी की लंबाई)



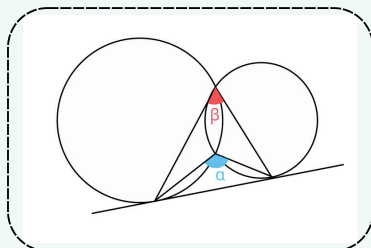
$2\pi r + 2d$



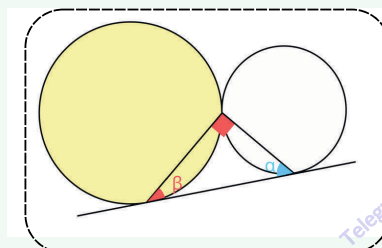
$2\pi r + 3d$



$2\pi r + 4d$



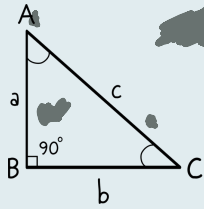
$$\alpha + \beta = 180^\circ$$



$$\alpha + \beta = 90^\circ$$

TRIANGLES (त्रिभुज)

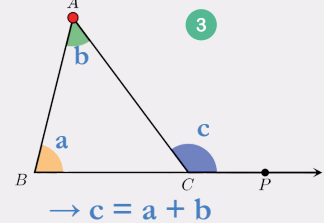
1 BASIC PROPERTIES (बुनियादी गुण)



$$|b - c| < a < b + c$$

- Sum of two sides always greater than third side (दो भुजाओं का योग हमेशा तीसरी भुजा से अधिक होता है।)
- Difference of two sides always less than third side (दो भुजाओं का अंतर हमेशा तीसरी भुजा से कम होता है।)

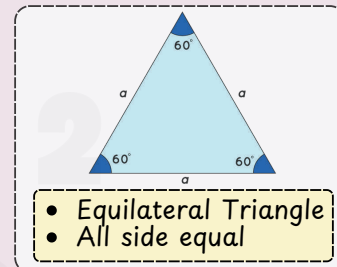
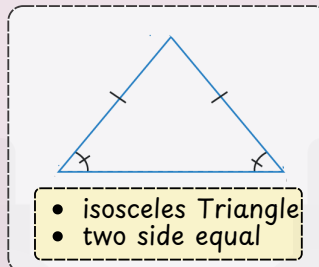
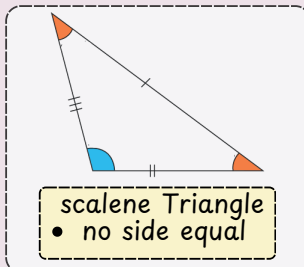
EXTERIOR ANGLE PROPERTY



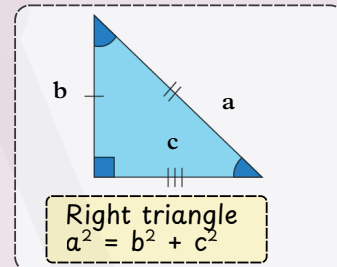
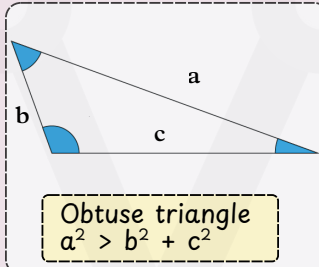
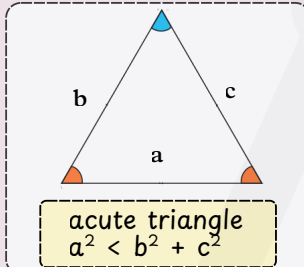
2 SUM OF ANGLES (कोणों का योग) $\rightarrow a + b + c = 180^\circ$

4 TYPES OF TRIANGLE (त्रिभुज के प्रकार)

- Based On Sides (भुजाओं के आधार पर)

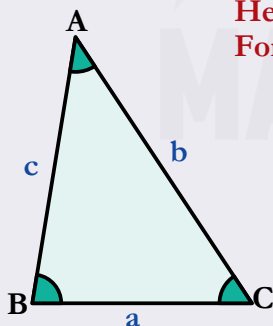


- Based On Angles (कोणों के आधार पर)



5 BASIC FORMULAS (मूलभूत सूत्र)

(विषमबाहु त्रिभुज के लिए)
For scalene triangle



Heron's
Formula

$$\text{Perimeter} = a + b + c$$

$$\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$$

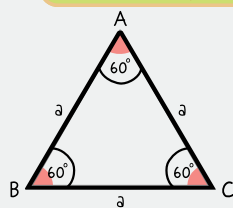
where 's' is semi perimeter $= \frac{a+b+c}{2}$

$$\text{Area} = \frac{1}{2} \times \text{Base} \times \text{height}$$

$$\text{Area} = \frac{1}{2} bc \sin A = \frac{1}{2} ab \sin C = \frac{1}{2} ac \sin B$$

EQUILATERAL TRIANGLE

(समबाहु त्रिभुज)



$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

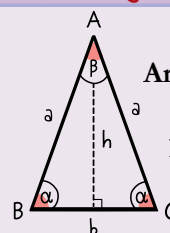
median is angle bisector

$$\text{Altitude (height)} = \frac{\sqrt{3}}{2} a$$

divides in two equal parts

ISOSCELES TRIANGLE

(समद्विबाहु त्रिभुज)



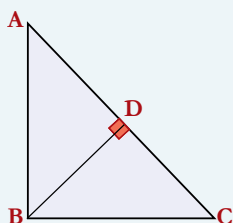
$$\text{Area} = \frac{b}{4} \sqrt{4a^2 - b^2}$$

$$h = \sqrt{a^2 - \frac{b^2}{4}}$$

- altitude from common vertices of two equal sides divides it into two equal parts
(दो बराबर भुजाओं के कॉमन वर्टिक्स से ऊंचाई इसे दो बराबर हिस्सों में बांटती है)

RIGHT ANGLE TRIANGLE

(समकोण त्रिभुज)

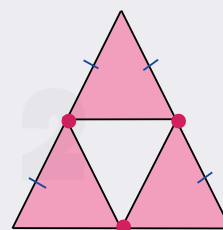


$$\begin{aligned} \triangle ABC &\sim \triangle ADB \\ \triangle ABC &\sim \triangle BDC \\ \triangle ADB &\sim \triangle CDB \end{aligned}$$

$$\begin{aligned} AB^2 &= AD \times AC \\ BC^2 &= CD \times CA \\ BD^2 &= AD \times DC \\ \frac{1}{BD^2} &= \frac{1}{AB^2} + \frac{1}{AC^2} \end{aligned}$$

$$BD \times AC = AB \times BC$$

$$\frac{AB^2}{BC^2} = \frac{AD}{DC}$$



- 4 equal parts
- all parts similar to original one & each other

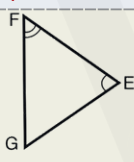
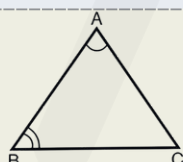
SIMILARITY & CONGRUENCY

(समरूपता और सर्वांगसमता)

Angle-Angle (AA)

If $\angle CAB \sim \angle GEF$ &
 $\angle ABC \sim \angle EFG$, then

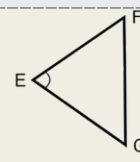
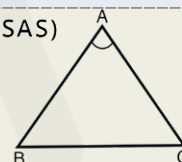
$$\triangle ABC \sim \triangle EFG$$



Side-Angle-Side (SAS)

If $\angle CAB \sim \angle FEG$ &
 $\frac{AB}{EG} = \frac{AC}{EF}$, then

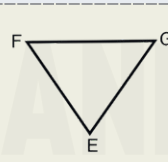
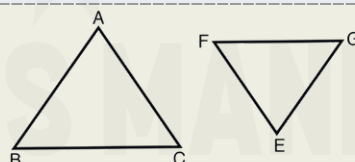
$$\triangle ABC \sim \triangle EFG$$



Side-Side-Side (SSS)

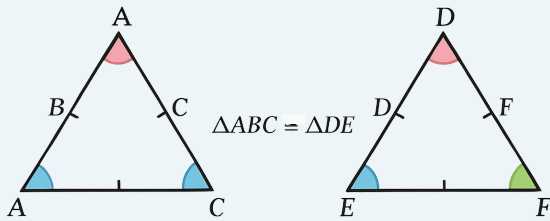
If $\frac{AB}{EF} = \frac{BC}{FG} = \frac{AC}{EG}$

$$\triangle ABC \sim \triangle EFG$$



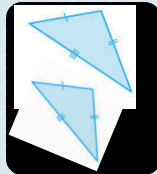
$$\frac{AB}{EF} = \frac{AC}{EG} = \frac{BC}{FG} = \frac{\text{med}(ABC)}{\text{med}(EFG)} = \frac{\text{angle bisector}}{\text{angle bisector}} = \frac{\text{perimeter}(ABC)}{\text{perimeter}(EFG)}$$

$$\frac{\text{Area}(ABC)}{\text{Area}(EFG)} = \frac{AB^2}{EF^2} = \frac{AC^2}{EG^2} = \frac{BC^2}{FG^2} = \frac{\text{med}(ABC)^2}{\text{med}(EFG)^2} = \frac{\text{angle bisector}^2}{\text{angle bisector}^2} = \frac{\text{perimeter}(ABC)^2}{\text{perimeter}(EFG)^2}$$



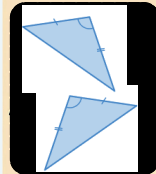
SSS

If all three sides of one triangle are equal to all three sides of another triangle



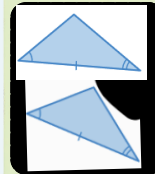
SAS

If two sides and the included angle of one triangle are equal to two sides and the included angle of another triangle



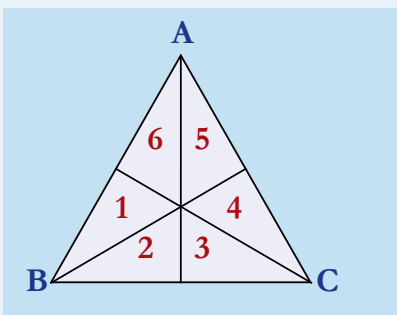
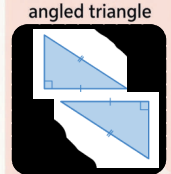
ASA

If two angles and the included side of one triangle are equal to two angles and the included side of another triangle



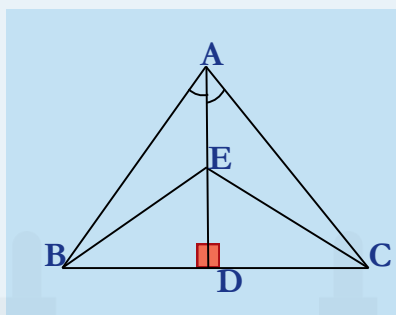
RHS

If the hypotenuse and one side of a right-angled triangle are equal to the hypotenuse and one side of another right-angled triangle

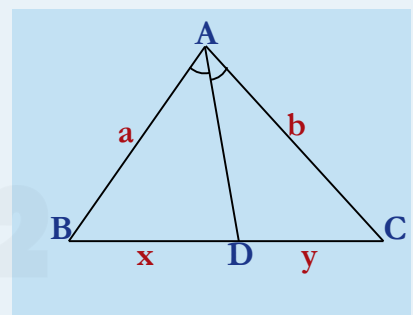


medians divides in 6 equal parts (for all triangles)

माधिकाएँ (सभी त्रिभुजों के लिए) त्रिभुज को 6 बराबर भागों में बाँटती हैं।

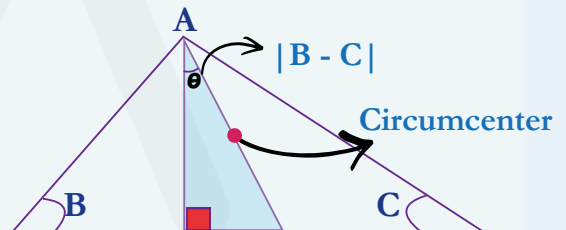
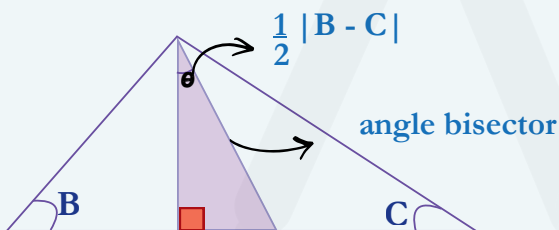


$$BE^2 + AC^2 = CE^2 + AB^2$$

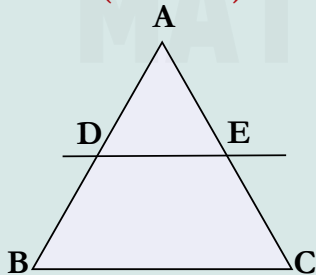


$$\frac{a}{b} = \frac{x}{y}$$

$$AD^2 = ab - xy$$

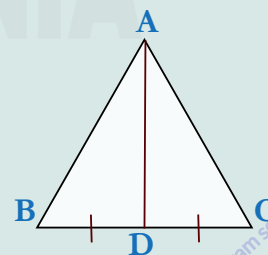


THALES THEOREM (थेल्स प्रमेय)



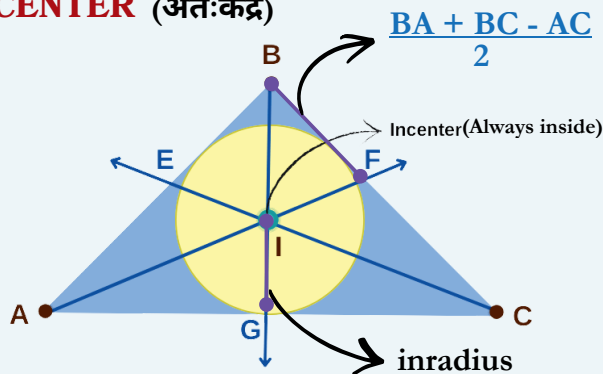
$$\frac{AD}{AB} = \frac{AE}{AC} \quad \text{OR} \quad \frac{AD}{DB} = \frac{AE}{EC} \quad \text{OR} \quad \frac{BD}{BA} = \frac{EC}{CA}$$

APOLLONIUS THEOREM (अपोलोनियस प्रमेय)



$$AB^2 + AC^2 = 2(AD^2 + BD^2)$$

INCENTER (अंतःकेंद्र)

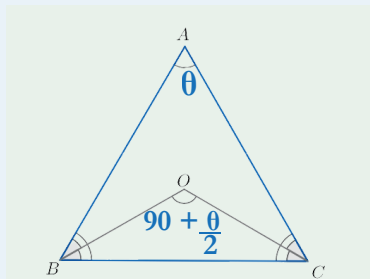


• **General Formula** = $\frac{\text{Area}}{\text{Semi-perimeter}}$
(सामान्य सूत्र)

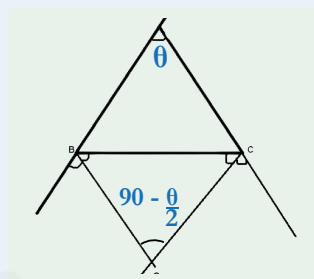
• **For Equilateral Triangle** = $\frac{a}{2\sqrt{3}}$
(अंतर्वृत्त की त्रिज्या)

• **For Right Angle Triangle** = $\frac{P + B - H}{2}$
(समकोण त्रिभुज के लिए)

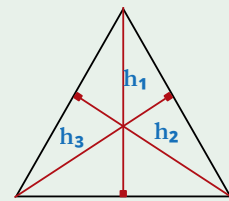
• $\frac{BI}{IG} = \frac{AB + BC}{AC}$, $\frac{AI}{IE} = \frac{AB + AC}{BC}$



INTERNAL ANGLE BISECTOR
(आंतरिक कोण समद्विभाजक)

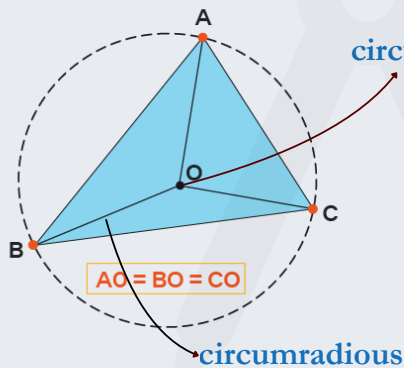


EXTERNAL ANGLE BISECTOR
(बाह्य कोण समद्विभाजक)



$$\frac{1}{r} = \frac{1}{h_1} + \frac{1}{h_2} + \frac{1}{h_3}$$

EXTERNAL ANGLE BISECTOR (बाह्य कोण समद्विभाजक)



circumcenter

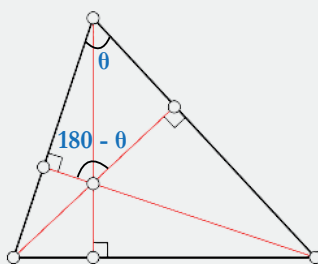
- For acute → inside
- For obtuse → outside
- For right → mid point of hypotenuse

• **General Formula** → $\frac{abc}{4 \times \text{area}}$
(सामान्य सूत्र)

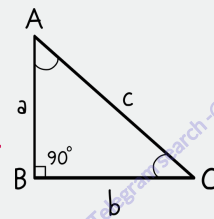
• **For Equilateral Triangle** → $\frac{a}{\sqrt{3}}$
(बाह्य वृत्त की त्रिज्या)

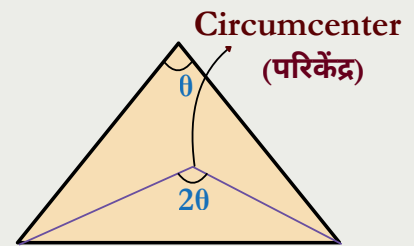
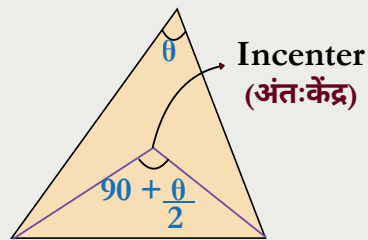
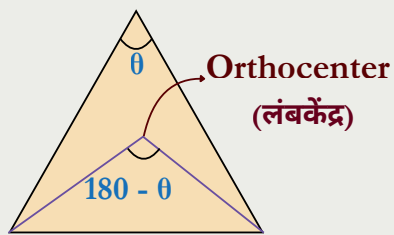
• **For Right Angle Triangle** → $\frac{H}{2}$
(बाह्य वृत्त की त्रिज्या)

• ORTHOCENTER (लंबकेंद्र)

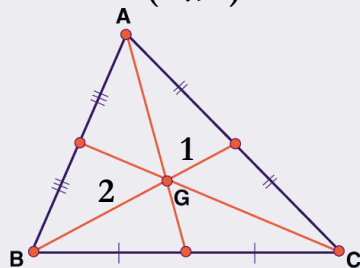


- acute → inside
- obtuse → outside
- right → vertices → 90°

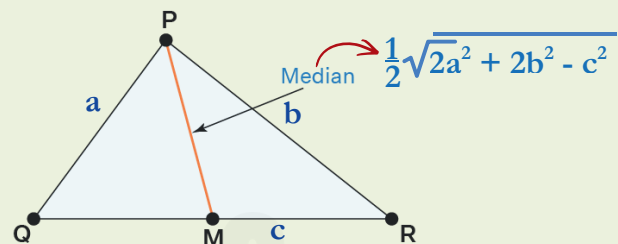




CENTROID (केंद्रक)



LENGTH OF A MEDIAN (माधिका की लंबाई)



EXTENDED SINE RULE (विस्तारित साइन नियम)

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

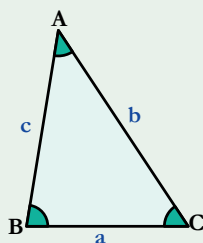
Circumradius

Distance between incenter
& circumcenter = $\sqrt{R^2 - 2Rr}$

(अंतःकेंद्र और परिकेंद्र
के बीच की दूरी)

- R = circumradius
- r = inradius

Cosine Rule (कोसाइन नियम)



$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

MENSURATION - 2D (क्षेत्रमिति - 2D)

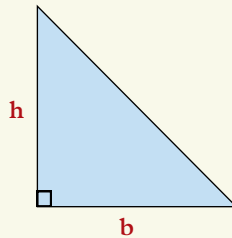
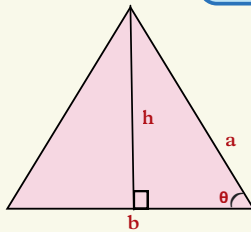
TRIANGLE (त्रिभुज)

AREA

$$\frac{1}{2} \times b \times h$$

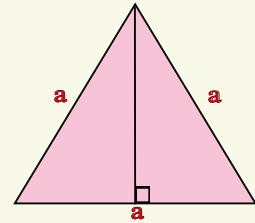
Heron's formula :- $\sqrt{s(s-a)(s-b)(s-c)}$
 $s = \frac{a+b+c}{2}$

$$\frac{1}{2} ab \sin \theta$$



for Right-Angled Triangle :-

$$\text{Area} :- \frac{1}{2} \times b \times h$$



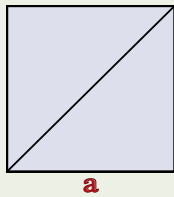
for Equilateral Triangle :-

$$\text{Altitude} :- \frac{\sqrt{3} a}{2}$$

$$\text{Perimeter} :- 3a$$

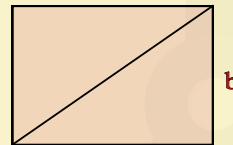
$$\text{Area} :- \frac{\sqrt{3} a^2}{4}$$

SQUARE (वर्ग)



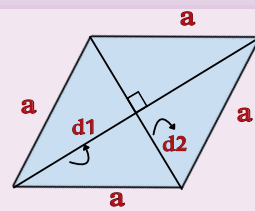
- Area :- a^2
- Perimeter :- $4a$
- Diagonal :- $\sqrt{2}a$
- if diagonal :- a
then side :- $\frac{a}{\sqrt{2}}$

RECTANGLE (आयत)



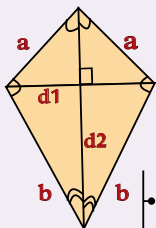
- Area :- $l \times b$
- Perimeter :- $2(l+b)$
- Diagonal :- $\sqrt{l^2 + b^2}$

RHOMBUS (समचतुर्भुज)



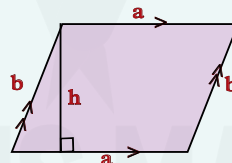
- Area :- $\frac{1}{2} \times d1 \times d2$
- Perimeter :- $4a$

KITE (पतंग)



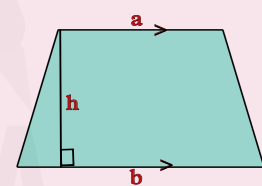
- Area :- $\frac{1}{2} \times d1 \times d2$
- Perimeter :- $2(a+b)$

PARALLELOGRAM (चतुर्भुज)



- Area :- $a \times h$
- Perimeter :- $2(a+b)$

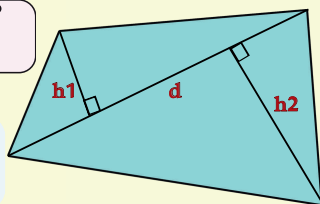
TRAPEZIUM (समलंब चतुर्भुज)



$$\text{Area} :- \frac{1}{2} \times (a+b) \times h$$

Area of any Quadrilateral
(किसी भी चतुर्भुज का क्षेत्रफल)

$$\text{Area} :- \frac{1}{2} \times d \times (h1+h2)$$

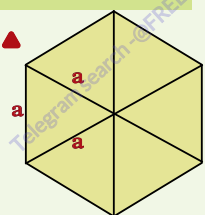


HEXAGON (षट्भुज)

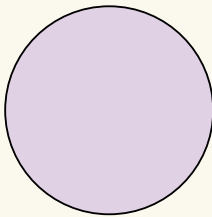
- Area :- $6 \times \text{ar}(\text{equilateral})$

$$6 \times \frac{\sqrt{3} a^2}{4}$$

- Perimeter :- $6a$



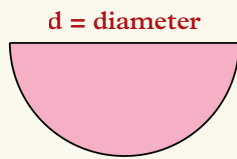
CIRCLE



- Area :- πr^2
- Circumference :- $2\pi r$

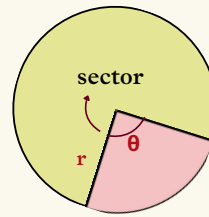
Distance covered in one complete revolution of a wheel equals its circumference.

SEMI-CIRCLE



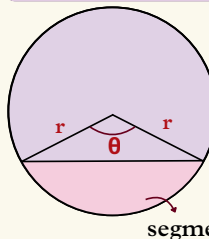
- Area :- $\frac{\pi r^2}{2}$
- Perimeter :- $\pi r + d$

SECTOR



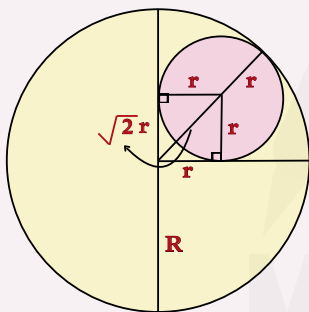
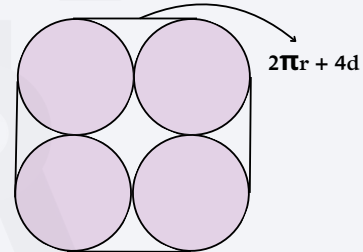
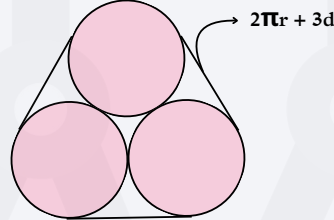
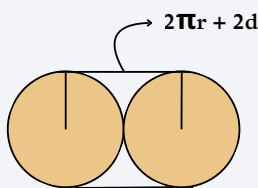
- Area :- $\frac{\theta}{360^\circ} \times \pi r^2$
- Perimeter :- $2r + \frac{\theta}{360^\circ} \times 2\pi r$
arc length

SEGMENT

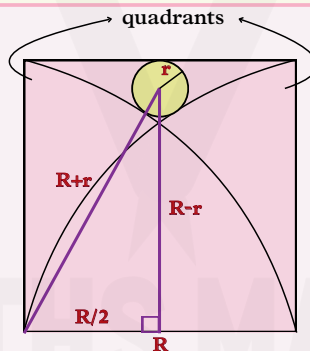


- Area :- ar of sector - ar of Δ
 $\frac{\theta}{360^\circ} \times \pi r^2 - \frac{1}{2} r^2 \sin \theta$

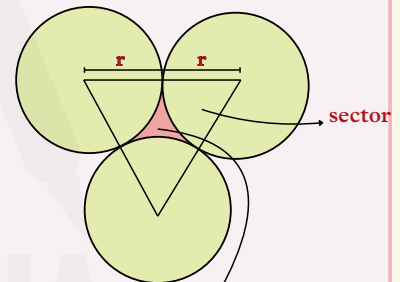
LENGTH OF STRING WRAPPED AROUND EQUAL CIRCLES (बराबर वृत्तों के चारों ओर लिपटे धागे की लंबाई)



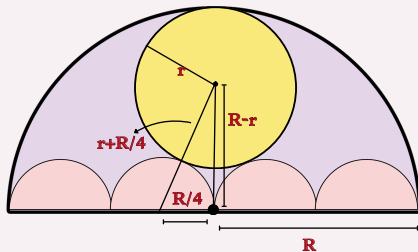
$$R = \sqrt{2} r + r$$



$$(R+r)^2 = (R-r)^2 + (R/2)^2$$

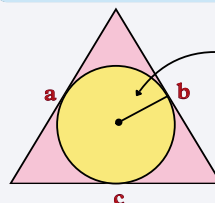


$$\frac{\sqrt{3}(2r)^2}{4} - \frac{1}{2} \pi r^2$$



$$(r+R/4)^2 = (R-r)^2 + (R/4)^2$$

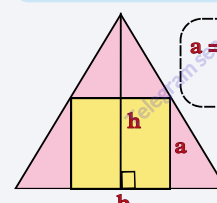
Circle of max. area inside



$$r = \frac{\text{ar}(\Delta)}{s}$$

$$s = \frac{a+b+c}{2}$$

Square of max. area inside



$$a = \frac{b \times h}{b + h}$$